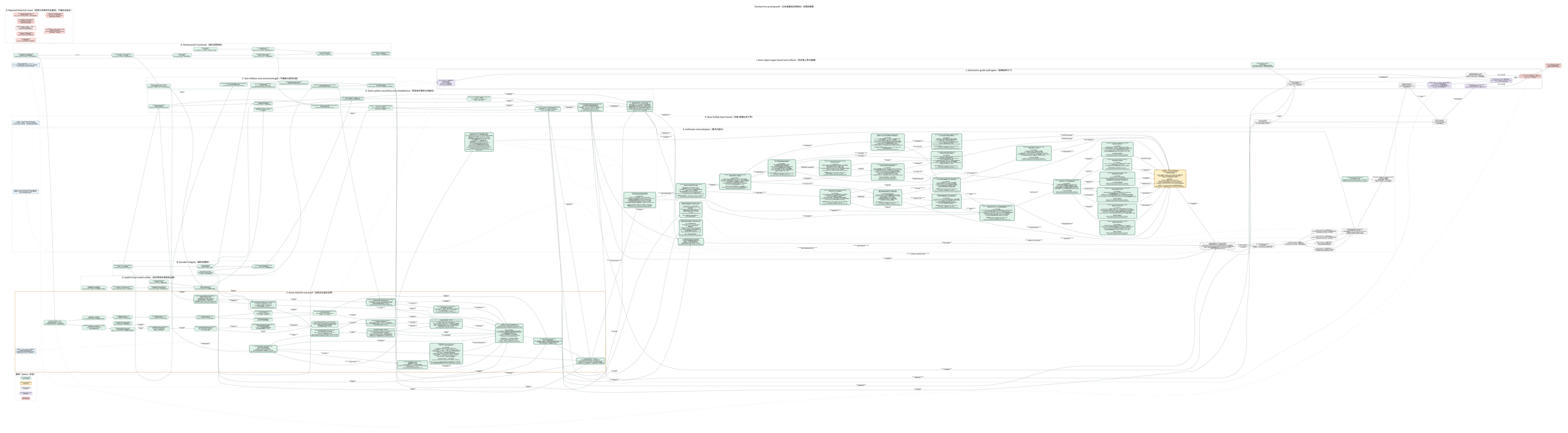

Checked min-proof growth（已检查最短证明增长）定理依赖图

K. Rejected historical routes（经审计弃用的历史路线；不通向主结论）
A. Honest proof coordinate（诚实证明坐标）
binaryNatCode / decodeBinaryNat
binaryTermCode + binaryFormulaCode
binarySequentCode
PAAxiomCertificate
StructuralValidityCertificate
CheckedPAProofTree binaryCode
binaryCertifiedPAProofCode
packBinaryString / packedPayloadLength
C. Non-inflation and canonical length（不膨胀与规范长度）
E. Same-system soundness and completeness（同系统可靠性与完备性）
decodeCompactListedProof
listedCompactCertifiedPAProofVerifier_eq_true_iff
decodeCompactListedProof_canonicalBinaryCode_append
derivation_to_listedCompactCertifiedPAProofVerifier completeness
Concrete P(x,y) / ListedCompactCertifiedPAProofChecks
minListedCertifiedPAProofPayloadLength(formula)
P_canon(code, formulaCode) = canonical Bool verifier
#print axioms audit
I. Same-object upper bound and collision（同对象上界与碰撞）
J. Submission-grade audit gates（投稿级审计门）
Sondow checked certificates
Sondow same-object compiler
Pudlak 1986 Theorem 3.1
Forbidden project assumptions
Final dependency audit
H. Buss-Pudlak lower bound（布斯-普德拉克下界）
Buss 1994 Theorem 5
G. Arithmetic internalization（算术内部化）
Quantitative parameterized diagonalization
PA consistency in the standard model
Consistency contradiction + exponent extraction
Bounded proof-calculus assembly
Pudlak condition (i)
Pudlak condition (ii)
Pudlak condition (iii)
suitable numeration
PA computation-trace theorem
B. Decoder integrity（解码完整性）
decodeCompactTerm / Formula / Sequent
decodeCompactProof
decodePAAxiomCertificate
decodeCompactPackedCertifiedPAProof
D. Explicit local costed verifier（显式带成本局部验证器）
bitStringEqTrace / natEqTrace
bitStringMemTrace / SubsetTrace / SetEqTrace
formulaEq/Mem/Subset/SetEqTrace
ListedCheckedPAProofTree
listedCertificateValidTrace
F. Active total bit-cost proof（当前总位成本证明）
decodeBinaryNatTrace_result
OptionTrace + traceBind
decodeManyVectorTrace_result
decodeCompactTermTrace_result
decodeCompactFormulaTrace_result
transformation binary-code length bounds
Success/failure total potential bound
图例（status / 状态）
Lean kernel checked
Lean 内核已检查
Active proof obligation
当前证明义务
Pending root theorem
待完成定理
Axiom/dependency audit
公理/依赖审计
Rejected route
经审计弃用路线